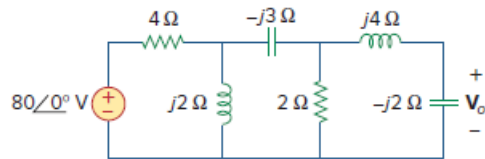


Problem

Use the concept of source transformation to find V_o in the circuit of Fig. 10.97.

Figure 10.97:



Step-by-step solution

Step 1 of 8

Refer to Figure 10.97 in the textbook.

Convert the voltage source into current source using source transformation.

$$I_s = \frac{80\angle 0^\circ}{4}$$

$$= 20\angle 0^\circ \text{ A}$$

Draw the modified circuit.

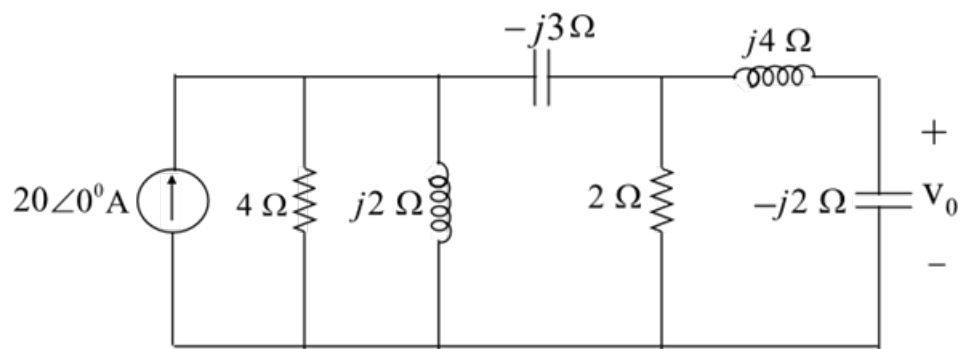


Figure 1

Step 2 of 8

Calculate the parallel combination of 4Ω resistance and $j2\Omega$ reactance.

$$Z_1 = \frac{4 \times j2}{4 + j2}$$

$$= \frac{j8}{4 + j2}$$

$$= 0.8 + j1.6 \Omega$$

Convert the current source into a voltage source using source transformation.

$$V_s = I_s Z_1$$

$$= (20\angle 0^\circ)(0.8 + j1.6)$$

$$= (20\angle 0^\circ)(1.79\angle 63.43^\circ)$$

$$= 35.78\angle 63.43^\circ \text{ V}$$

Step 3 of 8

Draw the modified circuit of Figure 1.

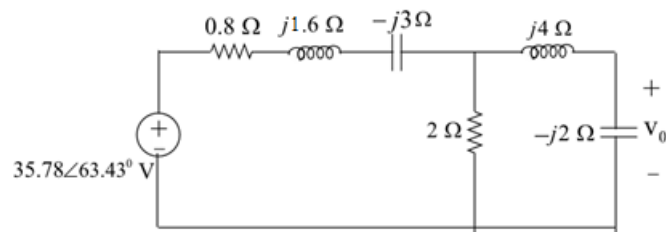


Figure 2

Step 4 of 8

Calculate the series combination of $0.8 + j1.6 \Omega$ impedance and $-j3 \Omega$ reactance.

$$\begin{aligned} Z_2 &= 0.8 + j1.6 - j3 \\ &= (0.8 - j1.4) \Omega \end{aligned}$$

Convert the voltage source into current source using source transformation.

$$\begin{aligned} I_{s1} &= \frac{V_s}{Z_2} \\ &= \frac{35.78 \angle 63.43^\circ}{0.8 - j1.4} \\ &= \frac{35.78 \angle 63.43^\circ}{1.61 \angle -60.26^\circ} \\ &= 22.19 \angle 123.69^\circ \text{ A} \end{aligned}$$

Step 5 of 8

Draw the modified circuit of Figure 2.

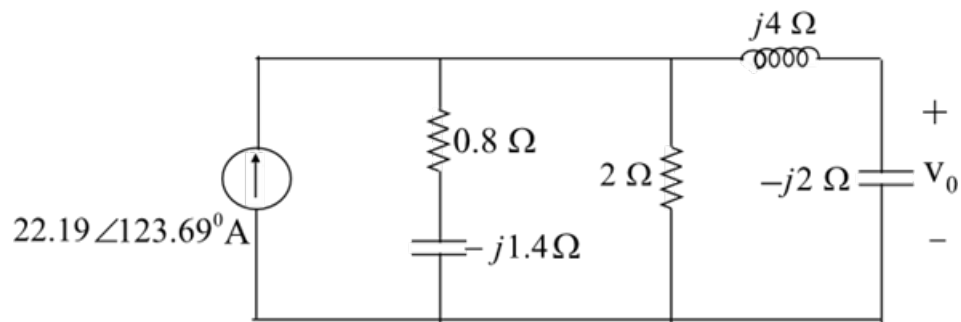


Figure 3

Step 6 of 8

Calculate the parallel combination of $0.8 - j1.4 \Omega$ impedance and 2Ω resistance.

$$\begin{aligned} Z_3 &= \frac{(0.8 - j1.4) \times 2}{0.8 - j1.4 + 2} \\ &= \frac{1.6 - j2.8}{2.8 - j1.4} \\ &= 0.86 - j0.571 \Omega \end{aligned}$$

Step 7 of 8

Convert the current source into a voltage source using source transformation.

$$\begin{aligned} V_{s1} &= I_{s1} Z_3 \\ &= (22.19 \angle 123.69^\circ)(0.86 - j0.571) \\ &= (22.19 \angle 123.69^\circ)(1.03 \angle -33.58^\circ) \\ &= 22.91 \angle 90.1^\circ \text{ V} \end{aligned}$$

Draw the modified circuit of Figure 3.

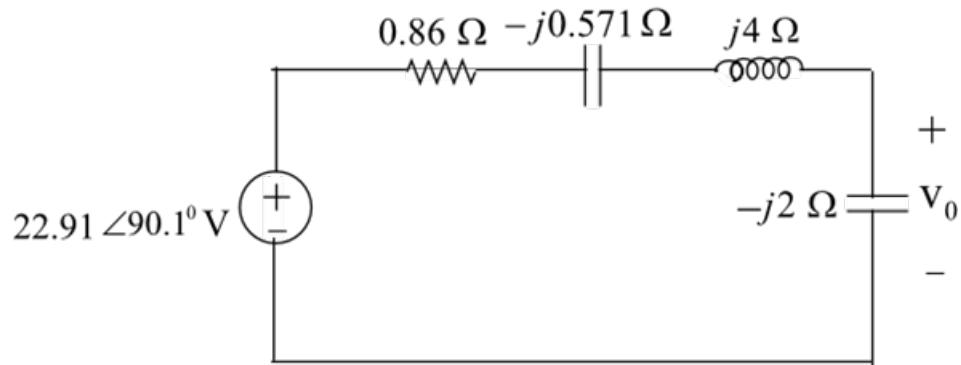


Figure 4

Step 8 of 8

Calculate the voltage v_o using voltage division rule.

$$\begin{aligned} v_o &= (22.91 \angle 90.1^\circ) \left(\frac{-j2}{0.86 - j0.571 + j4 - j2} \right) \\ &= (22.91 \angle 90.1^\circ) \left(\frac{-j2}{0.86 + j1.429} \right) \\ &= (22.91 \angle 90.1^\circ) \times \left(\frac{2 \angle -90^\circ}{1.668 \angle 58.96^\circ} \right) \\ &= 27.47 \angle -58.9^\circ \text{ V} \end{aligned}$$

Therefore, the voltage, v_o is $\boxed{27.47 \angle -58.9^\circ \text{ V}}$.